```
import pandas as pd
```

```
df= pd.read_excel('/content/Sales Dataset.xlsx')
```

```
df
```

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
0	1001	2024-10-01	Priya	Hyderabad	East	Mouse	Accessories	1735	7	0.00	307.52
1	1002	2024-05-03	Sneha	Kolkata	East	Monitor	Electronics	807	2	0.15	49.89
2	1003	2024-01-26	Anita	Mumbai	West	Phone	Electronics	2328	1	0.20	66.47
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
4	1005	2024-10-15	Rahul	Bangalore	South	Monitor	Electronics	3550	2	0.20	146.30
...	...	...	...	...	...	...	...	...	...	...	...
225	1226	2024-06-14	Anita	Ahmedabad	East	Laptop	Electronics	5647	2	0.15	662.00
226	1227	2024-04-02	Dev	Pune	South	Mouse	Accessories	1669	6	0.05	323.13
227	1228	2024-01-14	Vikas	Chennai	North	Tablet	Electronics	2682	7	0.15	119.82
228	1229	2024-09-04	Priya	Pune	East	Tablet	Electronics	2635	8	0.00	531.76
229	1230	2024-04-13	Meera	Kolkata	North	Monitor	Electronics	5087	8	0.10	416.72

230 rows × 11 columns

```
#Q1 Dataset Exploration

print(df.head(11))
print(df.info())
print(df.shape)
print(df.dtypes)
```

```
    Order_ID  Order_Date Customer       City Region   Product     Category  \
0       1001  2024-10-01    Priya  Hyderabad   East     Mouse  Accessories
1       1002  2024-05-03    Sneha    Kolkata   East   Monitor  Electronics
2       1003  2024-01-26    Anita     Mumbai   West     Phone  Electronics
3       1004  2024-06-06     Sara       Pune   East  Keyboard  Accessories
4       1005  2024-10-15    Rahul  Bangalore  South   Monitor  Electronics
5       1006  2024-07-04    Pooja  Bangalore   West   Printer       Office
6       1007  2024-06-24      Dev    Chennai   East   Monitor  Electronics
7       1008  2024-03-18    Kavya    Kolkata   East     Phone  Electronics
8       1009  2024-10-31    Kavya  Ahmedabad   East     Phone  Electronics
9       1010  2024-12-25    Pooja  Ahmedabad  North     Phone  Electronics
10      1011  2024-06-30     Neha      Delhi  South   Printer       Office

    Sales  Quantity  Discount   Profit
0    1735         7      0.00   307.52
1     807         2      0.15    49.89
2    2328         1      0.20    66.47
3    3933         3      0.20   -71.66
4    3550         2      0.20   146.30
5    3073         8      0.20   191.58
6    1170         5      0.20    22.83
7    4693         7      0.05   918.49
8    5071         6      0.10   707.55
9    2711         8      0.00   477.32
10   5977         6      0.00  1578.32
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 230 entries, 0 to 229
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order_ID    230 non-null    int64
 1   Order_Date  230 non-null    object
 2   Customer    230 non-null    object
 3   City        230 non-null    object
 4   Region      230 non-null    object
 5   Product     230 non-null    object
 6   Category    230 non-null    object
 7   Sales       230 non-null    int64
 8   Quantity    230 non-null    int64
 9   Discount    230 non-null    float64
 10  Profit      230 non-null    float64
dtypes: float64(2), int64(3), object(6)
memory usage: 19.9+ KB
None
(230, 11)
Order_ID        int64
Order_Date     object
```

```
Customer       object
City           object
Region         object
Product        object
Category       object
Sales           int64
Quantity        int64
Discount      float64
Profit        float64
dtype: object
```

```
#Q2 Sales Filtering Analysis

#Orders where Sales > 3500
High_sales= df[df['Sales']>3500]
print(High_sales)

#Orders where Region = "East" and Sales > 2500
region_sales = df[(df['Region']== 'East') & df['Sales']>2500]
print(region_sales)
```

```
     Order_ID  Order_Date Customer       City Region   Product     Category  \
3        1004  2024-06-06     Sara       Pune   East  Keyboard  Accessories
4        1005  2024-10-15    Rahul  Bangalore  South   Monitor  Electronics
7        1008  2024-03-18    Kavya    Kolkata   East     Phone  Electronics
8        1009  2024-10-31    Kavya  Ahmedabad   East     Phone  Electronics
10       1011  2024-06-30     Neha      Delhi  South   Printer       Office
..        ...         ...      ...        ...    ...       ...          ...
219      1220  2024-03-22    Meera  Bangalore   West     Phone  Electronics
222      1223  2024-03-04    Meera  Ahmedabad  South     Phone  Electronics
223      1224  2024-01-07      Dev  Bangalore   East    Tablet  Electronics
225      1226  2024-06-14    Anita  Ahmedabad   East    Laptop  Electronics
229      1230  2024-04-13    Meera    Kolkata  North   Monitor  Electronics

     Sales  Quantity  Discount   Profit
3     3933         3      0.20   -71.66
4     3550         2      0.20   146.30
7     4693         7      0.05   918.49
8     5071         6      0.10   707.55
10    5977         6      0.00  1578.32
..     ...       ...       ...      ...
219   5986         7      0.05   986.70
222   4356         6      0.10   416.87
223   4115         4      0.05   870.03
225   5647         2      0.15   662.00
229   5087         8      0.10   416.72

[104 rows x 11 columns]
Empty DataFrame
Columns: [Order_ID, Order_Date, Customer, City, Region, Product, Category, Sales, Quantity, Discount, Profit]
Index: []
```

```
#Q3 Customer Purchase Analysis
#Identify the top 5 customers based on total sales.
#Hint: Use GroupBy with aggregation.

Top_value = df.groupby('Customer')['Sales'].sum().sort_values(ascending=False).head(5)
print(Top_value)
```

```
Customer
Nitin    59820
Raj      58416
Rahul    52291
Dev      48500
Isha     46818
Name: Sales, dtype: int64
```

```
#Q4 Regional Sales Performance
#Calculate the total sales generated by each region.
#Then identify:
#Which region generated the highest revenue?

Region_sales = df.groupby('Region')['Sales'].sum().sort_values(ascending=False)
print(Region_sales)
```

```
Region
East     232675
South    183108
West     170227
North    158210
Name: Sales, dtype: int64
```

```
#Q5 Product Performance Analysis
#Calculate the total quantity sold for each product.
#Which product appears to be the most popular


total_Qty= df.groupby('Product')['Quantity'].sum().sort_values(ascending=False)
print(total_Qty)
```

```
Product
Monitor       156
Phone         151
Tablet        150
Laptop        140
Mouse         122
Headphones    115
Keyboard      110
Printer       108
Name: Quantity, dtype: int64
```

```
#Q6 Multi-Level Grouping
#Group the dataset by: Region and Category
#Calculate the total sales for each combination.
#What insights can you derive from this analysis

Region_category = df.groupby(['Region','Category'])['Sales'].sum().sort_values(ascending=False)
print(Region_category)
```

```
Region  Category
East    Electronics    160907
South   Electronics     86690
West    Electronics     84128
South   Accessories     80914
North   Electronics     70023
West    Accessories     64811
North   Accessories     62052
East    Accessories     57646
North   Office          26135
West    Office          21288
South   Office          15504
East    Office          14122
Name: Sales, dtype: int64
```

```
#Insight: Electronics dominates sales in every region, especially East. Office supplies sell the least everywhere.
```

```
#Q7 Pivot Table Creation
#Create a pivot table showing:
#Rows → Region, Columns → Product, Values → Total Sales
#Which region sells the most laptops?

pivot =pd.pivot_table(df,values='Sales',index='Region',columns='Product',aggfunc='sum',fill_value=0)
print(pivot)
```

```
Product  Headphones  Keyboard  Laptop  Monitor  Mouse  Phone  Printer  Tablet
Region
East          23899     10845   55568    31299  22902  39667    14122   34373
North         15450     20538    4540    14922  26064  30535    26135   20026
South         24111     23480   23029    19167  33323  18218    15504   26276
West          17159     38499   10826    18178   9153  27850    21288   27274
```

```
#Q8 Profitability Analysis
#Calculate the average profit for each product category.
#Which category is most profitable?



profit_category = df.groupby('Category')['Profit'].sum().sort_values(ascending=False)
print(profit_category)
```

```
Category
Electronics    48764.84
Accessories    33258.85
Office         12270.40
Name: Profit, dtype: float64
```

```
#Q9 Discount Impact Study
#Analyze the relationship between discount and profit.
#Tasks:
#Calculate the average profit for each discount level
#Identify whether higher discounts lead to lower profit
```

```
discount_profit = df.groupby('Discount')['Profit'].mean().sort_index()
print(discount_profit)

corr = df['Discount'].corr(df['Profit'])
print("Correlation:", corr)
```

```
Discount
0.00    751.210588
0.05    531.368039
0.10    346.460513
0.15    237.191400
0.20     90.026923
Name: Profit, dtype: float64
Correlation: -0.6637047529855844
```

```
#Q10 End-to-End Data Wrangling Task
#Perform the following workflow:
#Filter the dataset where Sales > 3000
#Group data by Region and Product
#Calculate total sales
#Reshape the result into a pivot table
#Explain the business insights obtained from this analysis.
```